

COMPOSITE PATTERN

Structural Design Pattern

The Problem: Managing Files and Folders

When you're building a system like a **file explorer** (with individual files and folders that can contain more files and folders), it becomes tricky to manage them because:

- Some items are individual files (like document.pdf)
- Some items are folders that contain other files and folders (like Photos/ containing pic1.jpg, pic2.jpg)

Treating files and folders differently everywhere in your code gets messy and complicated.